

Muhammad Hussain Ahmad

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SUMMARY

Robotics Control Engineer with 4+ years of experience across robotics and AI, delivering low-level control at the joint and actuator level on real hardware. Experienced in developing and deploying control systems on physical robots, from actuator-level torque control and state estimation to motion control, safety supervision, and perception-driven manipulation. Brings strong expertise in real-time C++, ROS 2/ros2_control, impedance and torque control, rigid-body dynamics, and robot safety.

EXPERIENCE

University of Manitoba

May 2024 – Present

Research Assistant, Robot Control & Safety

Winnipeg, Canada

- Shipped a **1 kHz real-time C++ torque controller** for a 7-DoF Kinova Gen3 with cyclic UDP and **lock-free** state, and implemented effort command handling and per-actuator torque **mode switching** in Kinova's **ros2_control** plugin.
- Resolved a **26.7 N·m** torque error **no simulation reproduced** by finding the Kinova API's torque sign reversed against the **URDF** and Pinocchio model, and traced a 198 ms torque-mode feedback gap to seven blocking mode-set round trips, proving it was not the fall cause with 0.07 N·m torque tracking.
- Built and tuned a **200 Hz Cartesian impedance controller** on hardware with gravity and friction **feedforward**, null-space regulation, barrier terms in joint and task space, and 12 Hz torque filtering.
- Designed the **actuator-level safety layer**, with torque saturation and slew limits inside the 1 kHz loop and a 200 Hz supervisor for E-stop, watchdog, and position-mode handover, and replaced a latched-fault path that free-fell the arm **7.45 degrees** with controlled **fault handling**.
- Added whole-arm self-collision avoidance over 26 URDF-mesh link pairs with barriers projected through **Jacobians** at **0.39 ms** p99.
- Identified payloads online with an **Extended Kalman Filter**, 0.96 kg at a 97 mm wrist offset, and predicted contact 198 ms ahead at **100% TPR** in a 10 ms budget with a digital twin.
- Held a 15-node 1 kHz to 100 Hz ROS 2 stack on schedule after the OS denied **SCHED_FIFO** by sequencing bring-up around a 250 ms **DDS discovery** stall and capping the watchdog below 0.25 s.
- Cut end-effector tracking error **57%** over 1,080 markerless camera teleoperation trials with a CasADi QP-based **numerical IK** retargeter at 2.59 ms p99 and zero missed deadlines.
- Developed a **browser-based** (HTML/JavaScript) **hardware debugging console** with live torque, tracking, fault, and mode-control telemetry.

Teleworker AI

Feb 2026 – Sep 2026

Robotics Engineer

Winnipeg, Canada

- Operated a ROS 2 scanning pipeline on a Unitree Go2 EDU's onboard Jetson Orin for daily 4-hour runs on a construction site, with a Leica BLK360 and Insta360 on a custom 3D-printed mount.
- Delivered Scan-to-BIM point clouds at **95%** export success by clearing robot, **sensor**, and embedded-compute failures on site.

Wombo

Aug 2024 – Dec 2024

Machine Learning Engineer

USA (Remote)

- Reduced SDXL inference latency from 2.5 s to 0.7 s and tripled throughput with LCM-LoRA, dynamic batching, and VRAM tuning, and cut Flux VRAM from 28 to 13 GB with BF16 and TensorRT.

Freelance

2022 – Jul 2024

Machine Learning Engineer

Remote

- Shipped a virtual staging pipeline with SDXL, ControlNet, and Depth Anything and cut its Docker builds from 2.5 min to 40 s.
- Lowered production Llama 3 8B inference latency 35% and p95 variance 40% under concurrency with INT8 quantization.
- Trained a Siamese face-verification CNN in TensorFlow with L1 distance scoring and a real-time OpenCV webcam verification pipeline.

PROJECTS

Load-Cell Force Sensing (B.S. Capstone)

Sensor design, bring-up, and calibration

- Designed and developed a 75 kg load cell, bonded its strain gauges, and **calibrated** it with Arduino, using SolidWorks and ANSYS FEA.

Gauntlet (open source)

Robot policy evaluation infrastructure

- Shipped continuous evaluation of learned and scripted policies across **MuJoCo**, PyBullet, Genesis, and Isaac Sim, gated by 3,678 **pytest** cases with ROS 2 publishing and Dockerized CI.

EDUCATION

University of Manitoba | M.Sc., Mechanical Engineering

May 2024 – Dec 2026 (Expected)

Thesis on real-time torque control, safety supervision, and shared autonomy on a 7-DoF manipulator

NUST | B.S., Mechanical Engineering

2019 – 2023

SKILLS

Control: impedance control, admittance control, torque control, PID, state feedback, observers, EKF, feedforward compensation, null-space control, control barrier functions, potential fields, filter design, system identification

Real-time and systems: real-time-safe object-oriented C++, lock-free programming, multithreading, RAIL, ROS 2 Jazzy, ros2_control, hardware interfaces, UDP, TCP/IP, DDS, real-time Linux, SCHED_FIFO, watchdogs, fault handling

Modeling and simulation: Pinocchio, CasADi, rigid-body dynamics, inverse kinematics, Jacobians, URDF, Xacro, MuJoCo, MJCF, Genesis, Isaac Sim, Gazebo, MATLAB

Sensors and perception: joint torque sensing, load cells, calibration, RealSense, MediaPipe, OpenCV, PyTorch, TensorRT, quantization

Tooling: CMake, colcon, Python, pytest, Docker, Git, HTML, JavaScript